

Nitin Dahiya

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Professional Summary

Product engineer who takes systems from idea to shipped reality — spanning backend architecture, AI integration, and applied security engineering. I build AI-integrated products end-to-end: AI Coach reasons over a user's real history instead of just chatting, and ThreatMap automates and triages vulnerability assessments with a fully local AI pipeline. I move fast by pairing strong system design with modern engineering practices.

Selected Projects

- AI Coach** — AI-Powered Coaching Platform (*building toward launch*) GitHub
- Designing and building an end-to-end AI coaching platform (Next.js App Router, TypeScript, Supabase Auth/PostgreSQL, Zod-validated APIs) that turns a user's real training history into personalized, explainable coaching decisions.
 - Architected the reasoning pipeline to separate deterministic computation from LLM-based reasoning: workout history → trend/progression analysis → LLM reasoning over computed facts → structured output, through a provider-agnostic LLM layer (Gemini, Groq) swappable without touching business logic.
 - Built database-backed memory (Supabase/PostgreSQL, Row-Level Security) as the single source of truth for user history, keeping recommendations reproducible and auditable; next iteration adds semantic retrieval (RAG, pgvector).
- forge-API** — Backend API System GitHub
- Built a modular backend API (FastAPI, PostgreSQL, SQLAlchemy 2.0, Alembic) with JWT/OAuth2 authentication, password hashing, and ownership-based authorization on protected endpoints.
 - Set up an automated test suite (pytest, pytest-cov, FastAPI TestClient/HTTPX) and a GitHub Actions CI pipeline that provisions PostgreSQL, runs migrations, and runs tests on every push; containerized with Docker Compose with automatic startup migrations.
- ThreatMap** — Open-Source Security Assessment Engine (*v1.0 shipped; v2: AI-vuln scanning*) GitHub
- Architected a plugin-based scanner engine (automatic plugin discovery, topological dependency ordering, YAML-configured profiles) orchestrating Nmap, Nuclei, Nikto, Gobuster, Subfinder, and SSLScan in parallel, **automating 90%+** of a manual assessment workflow; open-sourced publicly.
 - Added local AI-based triage — an on-device SLM that weighs severity, exploitability, and exposure context to rank findings — plus automated HTML/Excel/JSON reporting; adopted by other users.
- Win_Forensics** — Windows Digital Forensics Toolkit (*used in active investigations*) GitHub
- Reverse-engineered the WhatsApp Desktop client's encryption workflow — deriving DPAPI-wrapped keys and decrypting AES-OFB encrypted SQLite databases — to recover contacts, chats, and media from local storage.
 - Built a portable forensic parser producing structured JSON output, packaged as a standalone executable; deployed in real digital-forensics casework.
- Roast Arena** — Realtime Backend System (*in development, with another developer*)
- Built the backend (Fastify, TypeScript, PostgreSQL/Prisma, Redis Pub/Sub) for a realtime multiplayer web app, using native WebSockets to synchronize live battle state, submissions, and voting with idempotent event handling.
 - Implemented session-based auth (HTTP-only cookies), RBAC authorization, and a modular-monolith architecture (repository/service/route separation) with gamification (XP, badges, leaderboards); CI via GitHub Actions with Vitest coverage.

Experience

- Intern** — AbyM Technologies Feb 2026 – May 2026
Remote, Noida, India
- Assessed **7–8** internal web services as part of routine vulnerability reviews and identified the manual, repetitive scanning workflow as the real bottleneck.
 - Used that as the direct motivation for ThreatMap (see Projects) — an open-source automation pipeline now replacing workflows like this one end-to-end.
- Intern** — National Cyber Forensics Lab (IFSO), Delhi Police Jul 2025 – Oct 2025
Delhi, India
- Conducted digital forensic analysis on Windows endpoints — registry hives, event logs, and filesystem artifacts — to support active criminal investigations.
 - Built Python automation scripts that **automated 80%+** of forensic artifact collection and log analysis; **received a formal Appreciation Letter from Delhi Police** for the work.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Bash
Backend & APIs: FastAPI, Fastify, REST APIs, WebSockets, SQLAlchemy, Pydantic, Zod, Alembic
Databases: PostgreSQL, MySQL, Supabase (Auth, RLS), Prisma, Redis, pgvector
Testing & CI/CD: pytest, pytest-cov, HTTPX, FastAPI TestClient, Vitest, GitHub Actions
AI: Gemini, OpenAI, Groq, Anthropic APIs, RAG, Prompt Engineering, AI System Architecture
Frontend & DevOps: Next.js (App Router), React, Tailwind CSS, Docker, Docker Compose, Linux, Git, GitHub
Security: Security Automation, Vulnerability Assessment, RBAC/Authorization, Application Security, Digital Forensics

Education

Vivekananda Institute of Professional Studies, Delhi 2026
Bachelor of Technology in Computer Science & Engineering (Industrial IoT) — CGPA: 7.83/10